

Pain in 3D: Controllable Generation of Synthetic Faces for Automated Pain Assessment



Xin Lei Lin · Soroush Mehraban · Abhishek Moturu · Babak Taati

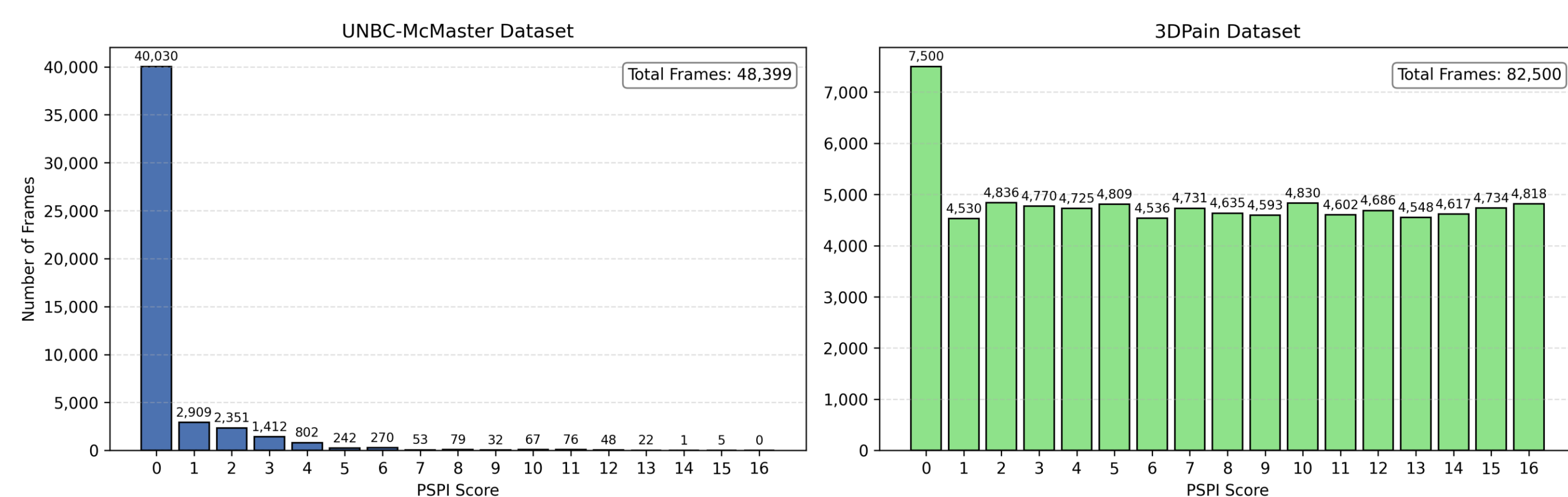
Vector Institute · KITE Research Institute (UHN) · University of Toronto



TAKE-HOME: Mesh-level control over facial Action Units yields clinically-valid, balanced synthetic pain data; closing the generalization gap left by scarce real datasets.

1 Motivation & Problem

- Automated pain assessment is vital for patients who cannot self-report (dementia, dementia, cognitive impairment).
- Existing datasets are tiny and demographically skewed — UNBC-McMaster has only 25 subjects.
- Prior generative models lack precise control over facial Action Units (AUs) and cannot guarantee clinical validity.



3DPain covers all PSPI levels uniformly, while real UNBC-McMaster frames collapse to near-zero pain — the imbalance ViTPain must overcome.

2 Contributions

- Controllable generation pipeline: FLAME mesh → depth-ControlNet face synthesis synthesis (Kandinsky 2.2) → PBR texturing (Hunyuan3D 2.1) → AU-based rigging → rigging → multi-view render → hair/background inpainting.
- 3DPain dataset: 82,500 frames, 25,000 pain expressions, 2,500 identities, balanced by age, gender, and ethnicity, with AU + PSPI + heatmap annotations.
- ViTPain: a Vision Transformer that cross-attends to a neutral reference face for identity-invariant pain scoring.

3 Dataset: 3DPain — Sample Identities

- 2,500 synthetic identities, each with paired neutral/pain images, AU + PSPI labels, and pain-region heatmaps — vs. 25 subjects in UNBC-McMaster.



Controllable AU-driven pain expressions across diverse identities, ages, and ethnicities.

4 Results — vs. State of the Art

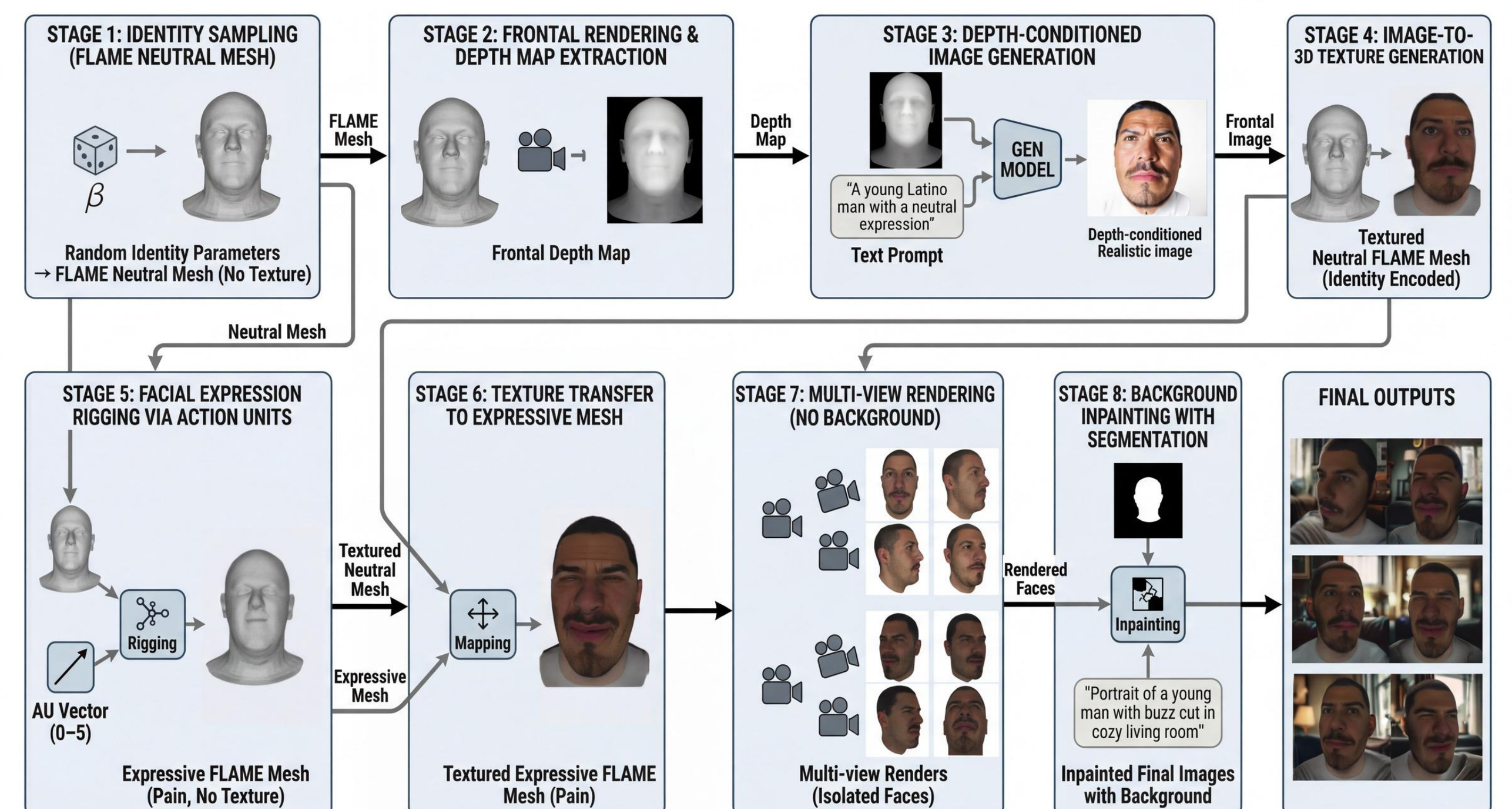
- 3DPain pretraining lifts every backbone: Vanilla ViT AUROC 0.54→0.73, ViTPain 0.75→0.86 (PSPI ≥ 1).
- From single frames, ViTPain matches video-based SOTA — tying the best F1 (0.59) and video ViViT's AUROC (0.86).

Method	Input	AUROC	F1
Rodriguez (CNN-LSTM) †	Video	.93	—
Rau (VSwin-T)	Video	.87	.59
Fiorentini (ViViT)	Video	.86	.55
Fiorentini (ViT)	Image	.88	.55
Rau (Swin)	Image	.80	.53
ViTPain (Ours)	Image	.86	.59

All PSPI ≥ 1, 5-fold CV; † LOSO protocol. ViTPain uses single frames — no temporal context.

5 Method

- The pipeline generates photorealistic, multi-view pain expressions by controlling the underlying 3D geometry.



Depth-conditioned FLAME synthesis (Kandinsky 2.2 + ControlNet), Hunyuan3D texturing, AU rigging, and rigging, and inpainting — stages labeled above.

$$PSPI = AU4 + \max(AU6, AU7) + \max(AU9, AU10) + AU43$$

Brow Lowerer

Cheek Raiser

Lid Tightener

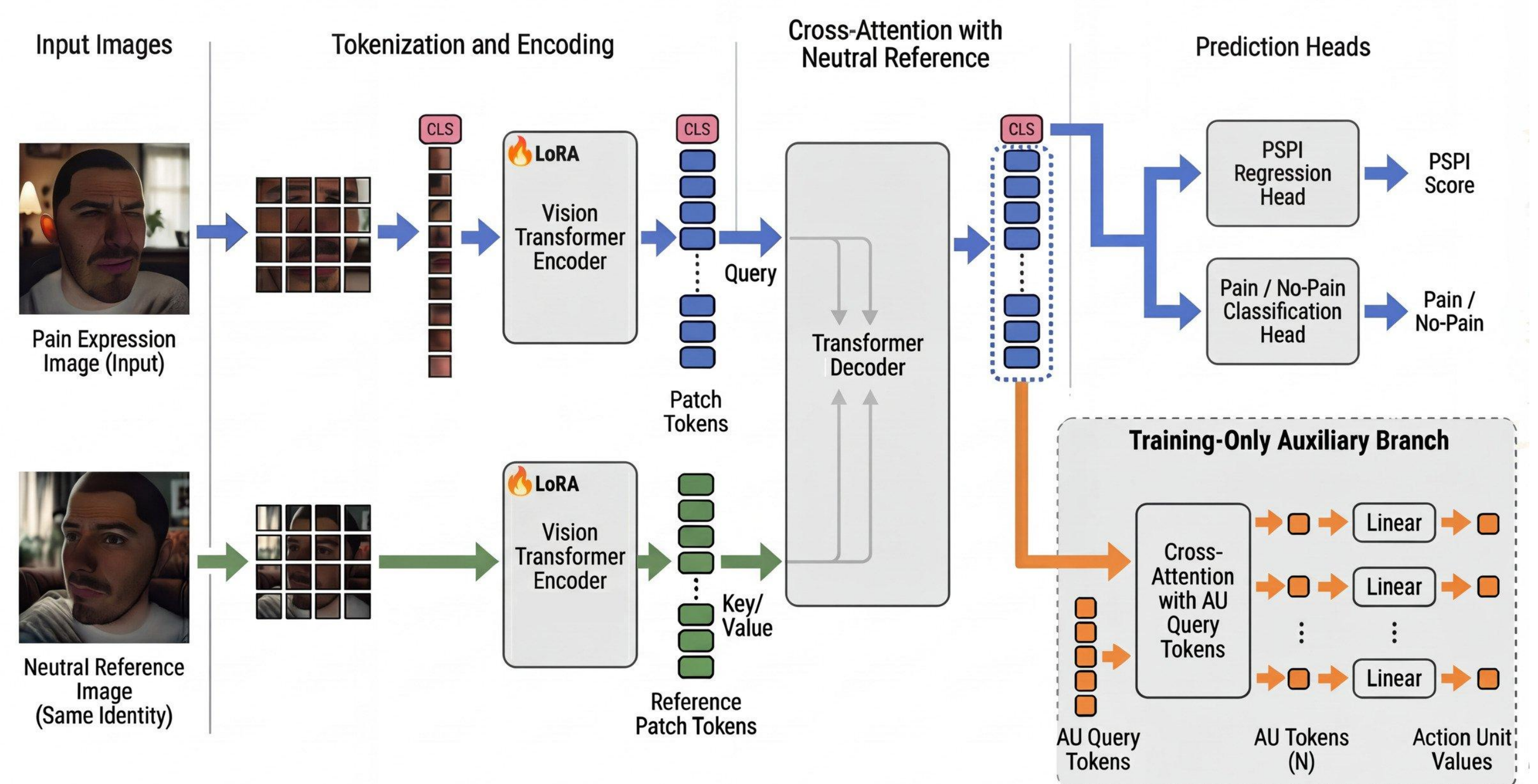
Nose Wrinkler

Upper Lip Raiser

Eyes Closed

6 ViTPain Architecture

- Pain image + neutral reference → DINOv3 encoders → cross-attention → PSPI PSPI regression + AU heads.



Cross-attention against a neutral reference makes pain scoring identity-invariant.

7 Conclusion & Future Work

- Precise AU-level control over the 3D facial mesh lets us synthesize clinically-clinically-valid, demographically-balanced pain expressions that are ethically ethically infeasible to collect from real patients.
- This controllability, not scale alone, is what transfers: 3DPain closes the generalization gap on real clinical faces despite the synthetic-to-real domain shift.
- Limitations: residual gap in skin-texture realism; no temporal modeling yet.
- Future work: higher photorealism and temporal consistency to further narrow the sim-to-real gap.

8. Acknowledgements

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